

Pablo Zúñiga Carvajal

Industrial & Systems Engineer (BSc & MSc Equivalent)

Data Modeling • Applied Statistics • Complex Systems

pablo.zuniga.c@gmail.com • +56 9 4941 9529 • pablozunigac.github.io

Santiago, Chile • Open to Relocation • Visa Eligibility: US H-1B1 & EU Blue Card

Tech Stack

- Data Engineering & Cloud: Python (FastAPI, Polars), TypeScript, SQL, MS Azure, ETL/ELT pipelines, Docker, GitHub Actions (CI/CD), relational databases, and Parquet.
- Advanced Analytics & Systems: R (Tidyverse, targets), inferential & predictive statistical modeling, Monte Carlo simulations, BPMN/UML, data governance, and Docs-as-Code knowledge architectures.

Work Experience

Cuxhaven Labs (*Proprietary R&D Studio*)

May 2026 – Present

Principal Systems Architect & Founder

Santiago, Chile

- Architecting ETL/ELT pipelines and semantic data ontologies to integrate heterogenous data sources.
- Systemic reduction of technical debt and refactoring system models backed by Docs-as-Code workflows.

Systems Engineering & Advanced Analytics Advisory

Jul 2022 – Present

Independent Forward Deployed Engineering Consultant

Valparaíso, Chile

- Engineered Python/R exponential statistical models optimizing GNSS pacing data across 1.8k+ triathlon athletes.
- Designed database structures, automated data flows, and Quarto/Markdown documentation architectures.

Decentralized Systems & Web3 Protocols

Jan 2020 – Jul 2022

Web3 Systems & On-Chain Infrastructure Engineer

Global / Remote

- Controlled DeFi assets via cold/non-custodial cross-chain wallets and DEXs networks.
- Coordinated NFT smart contract deployments for 1,100+ concurrent users in distributed environments.

Pontificia Universidad Católica de Valparaíso

Aug 2018 – Jun 2021

Academic Data Engineer & Technical Consultant

Valparaíso, Chile

- Engineered ETL pipelines for multi-source academic datasets; mentored STEM capstone & data science projects.

Banco de Crédito e Inversiones (BCI) (Data & Analytics Corporate VP)

May 2019 – Sep 2019

Process Engineer

Santiago, Chile

- Orchestrated metadata integration from core systems into MS Azure, standardizing BPMN process workflows.

Education

Pontificia Universidad Católica de Valparaíso

2011 – 2017

Industrial & Systems Engineering (Integrated 6-year program, BSc & MSc Equivalent)

Valparaíso, Chile

Core focus: systems theory, information technology, inferential statistics, operations research, and econometrics.

Projects & Internships

ESVAL – Lead Capstone Project | Viña del Mar, Chile

Jul 2017 – Dec 2017

Spatio-temporal predictive risk model: Cox Proportional Hazards in R (5,300 km of pipeline 1.9M+ residents).

ENAMI – Industrial Engineering Intern | Atacama & Santiago, Chile

Jan 2016 – Mar 2016

Audited metallurgical inventory database (75k+ SKUs), uncovering \$2.5M USD in accounting discrepancies.

Antofagasta Minerals – Geotechnical Engineering Intern | Salamanca, Chile

Jan 2014 – Mar 2014

Monitored open-pit mining geotechnical telemetry for slope failure prediction and early risk mitigation.

Technical Instruction

Pontificia Universidad Católica de Valparaíso – School of Business

2019

Graduate Technical Instructor

Valparaíso, Chile

BancoEstado / Universidad Federico Santa María

2017, 2019

Advanced Analytics Speaker & Trainer

Santiago, Chile

Pontificia Universidad Católica de Valparaíso – School of Industrial Engineering

2018

Guest Lecturer (Applied Data Science)

Valparaíso, Chile